Northeastern University, Khoury College of Computer Science

CS 6120 — Some Extra Practice 2

Due: February 23, 2026 (100 points)

In the last lecture, there are a few takeaways that would help you, and here are some interesting problem sets that could help you prepare.

Question 1: Trigram Token Merging

We define the following tokenization algorithm:

- Each word is represented as a sequence of characters.
- A special boundary symbol `|` marks word boundaries.
- The **initial vocabulary** consists of all unique characters in the corpus plus `|`.
- At each step:
  1. Count all adjacent **trigrams** (three consecutive symbols) that do not cross `|`.
  2. Merge the most frequent trigram into a single new token.
  3. Replace all occurrences of that trigram in the corpus.
- Break ties lexicographically.

Given Corpus

`banana|bandana|ban`

Initial character-level tokenization:

`b a n a n a | b a n d a n a | b a n`

Task

Continue performing trigram merges until the vocabulary size reaches **7**.

Hints and Walkthrough

Here are some helpful hints on how to think about this. Only look at this section after you've tried already! In order to solve this problem, a useful procedure would be the following.

1. List the initial vocabulary and its size.
2. Determine how many merges are required.
3. Perform the necessary merges.
4. Write the final tokenized corpus.
5. Write the complete final vocabulary in order of creation.

Question 1: Solutions

This is the final solution to the question. Please try the question before proceeding here.

Unique characters in the corpus:

$$\{a, b, d, n, |\}$$

Initial vocabulary size: 5
Target vocabulary size: 7

$$\text{Number of merges required} = 7 - 5 = 2$$

First Merge

Initial tokenization:

```
b a n a n a | b a n d a n a | b a n
```

Trigram Frequencies

Trigram	banana	bandana	ban
ban	1	1	1
ana	2	1	0
nan	1	0	0
and	0	1	0
nda	0	1	0
dan	0	1	0
Total			
ban		3	
ana		3	
others		1	

Tie between `ana` and `ban`.
Lexicographically: `ana` < `ban`

First merge: ana

Updated corpus:

`b ana n a | b a n d ana | b a n`

Second Merge

Trigram Frequencies After First Merge

Trigram	Count
b a n	2
b ana n	1
ana n a	1
a n d	1
n d ana	1

Second merge: ban

Updated corpus:

`b ana n a | ban d ana | ban`

Final Vocabulary

The final solution is as follows:

Original Tokens	{a, b, d, n, —}
New Tokens	ana, ban
Final Vocabulary Size	7
Final Vocabulary	$\boxed{\{a, b, d, n, \mid, ana, ban\}}$

Question 2: Edit Distance

Consider the source word **"kitten"** and the target word **"sitting"**. The allowed edit operations and their costs are:

Operation	**Cost**
Insertion	1
Deletion	1
Substitution	2

In this question, you must:

1. **Recurrence Relation:** Write the recurrence relation for $D(i, j)$
2. **Base Cases:** State the base cases for $D(i, 0)$ and $D(0, j)$
3. **Calculation:** Compute the value of $D(6, 7)$ to find the minimum edit distance between "kitten" and "sitting"

Question 2: Solutions and Walkthrough

1. **Recurrence Relation**

$$D(i,j) = \min \begin{cases} D(i-1,j)+1 & \text{(Deletion)} \\ D(i,j-1)+1 & \text{(Insertion)} \\ D(i-1,j-1)+\text{sub_cost} & \text{(Substitution)} \end{cases}$$

   Where sub_cost = 0 if source[i] = target[j], and sub_cost = 2 otherwise.

2. **Base Cases**
   - $D(i,0) = i$ (Deleting all i characters of the source)
   - $D(0,j) = j$ (Inserting all j characters of the target)

3. **3. Calculation for $D(6,7)$ (kitten → sitting):**
   - k → s: Substitute (2)
   - i → i: Match (0)
   - t → t: Match (0)
   - t → t: Match (0)
   - e → i: Substitute (2)
   - n → n: Match (0)
   - $\emptyset$ → g: Insert (1)

Total Distance: $2+0+0+0+2+0+1 = \mathbf{5}$

Question 3: Autocorrect Decision Rule

Suppose an autocorrect system observes the misspelled word: **"grafic"**. The system considers the following candidate corrections:

Candidate (w)	**Language Model Probability $P(w)$**
traffic	0.001
graphic	0.1
gratic	0.0001
graffiti	0.01

The system selects the correction w^* that **minimizes** the following score:

$$\text{Score} = \text{Edit Distance}(\text{"grafic"}, w) - \log_{10} P(w)$$

Tasks:

1. Compute the Edit Distance for each candidate (using costs: Ins=1, Del=1, Sub=2).
2. Compute the final score for each candidate.
3. Which word does the system choose as the correction?

Question 3: Solutions

The scoring formula is: Score = Edit Distance $- \log_{10} P(w)$.

Candidate (w)	**Edit Dist.**	$P(w)$	$-\log_{10} P(w)$	**Total Score**
traffic	3 (t, r, i)	10^{-3}	3	$3 + 3 = 6$
graphic	**1 (h)**	10^{-1}	**1**	$\mathbf{1 + 1 = 2}$
gratic	1 (t)	10^{-4}	4	$1 + 4 = 5$
graffiti	3 (f, i, i)	10^{-2}	2	$3 + 2 = 5$

The system chooses **"graphic"** because it minimizes the total score (Score = 2)